EXPERIMENT 8

BETWEEN, IN and Aggregate Functions

Aim: To filter records with BETWEEN and IN and calculate summaries using aggregate functions.

Software Required: MySQL 8.0 Command Line Client

SQL Program

```
USE SQL_Lab;

CREATE TABLE Exam_Marks (
    student_id INT PRIMARY KEY,
    student_name VARCHAR(50),
    department VARCHAR(20),
    marks INT
);

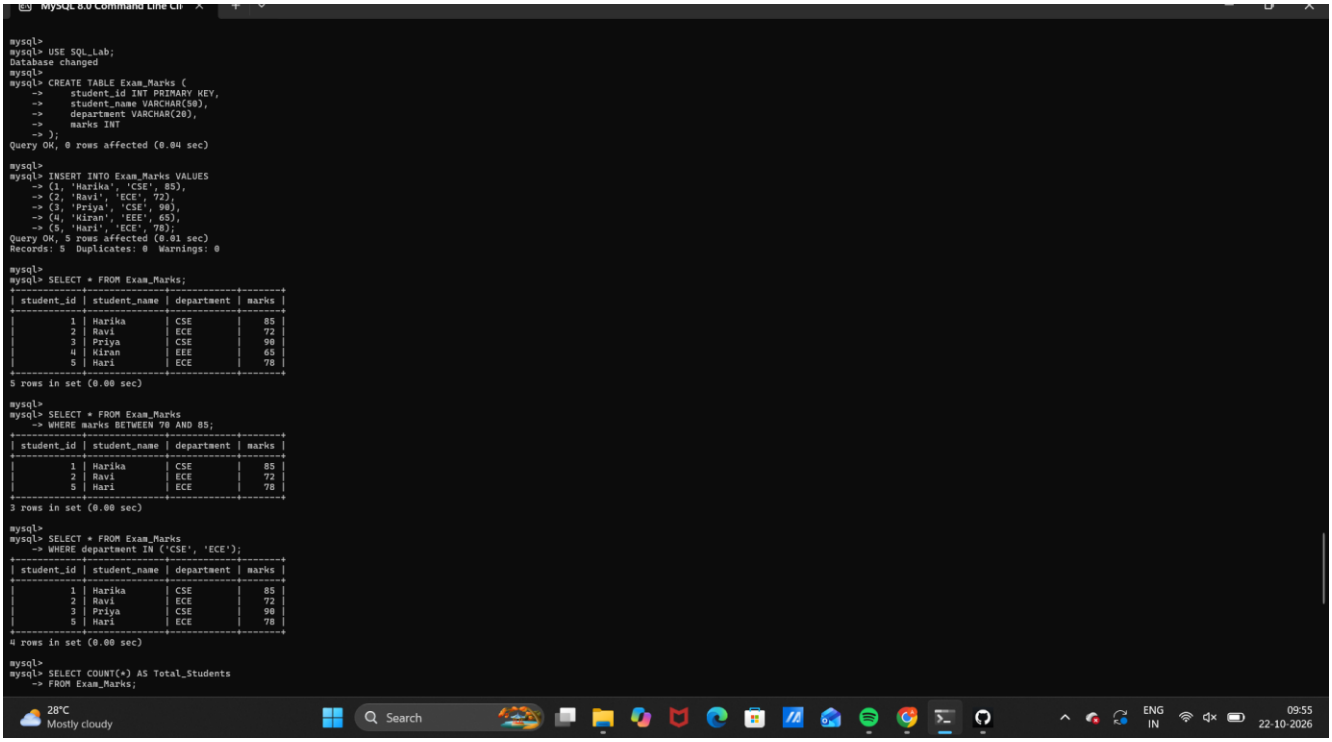
INSERT INTO Exam_Marks VALUES
(1, 'Harika', 'CSE', 85),
(2, 'Ravi', 'ECE', 72),
(3, 'Priya', 'CSE', 90),
(4, 'Kiran', 'EEE', 65),
(5, 'Hari', 'ECE', 78);

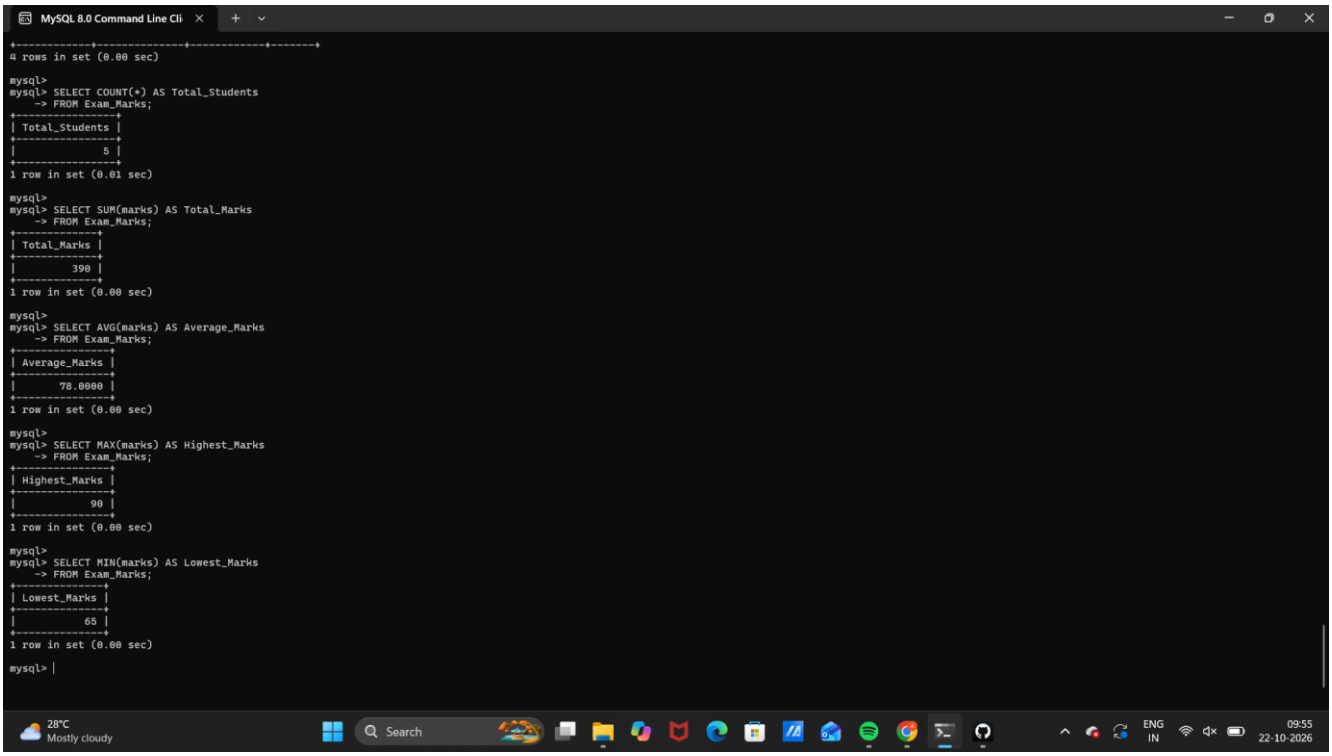
SELECT * FROM Exam_Marks;

SELECT * FROM Exam_Marks
WHERE marks BETWEEN 70 AND 85;

SELECT * FROM Exam_Marks
WHERE department IN ('CSE', 'ECE');

SELECT COUNT(*) AS Total_Students FROM Exam_Marks;
SELECT SUM(marks) AS Total_Marks FROM Exam_Marks;
SELECT AVG(marks) AS Average_Marks FROM Exam_Marks;
SELECT MAX(marks) AS Highest_Marks FROM Exam_Marks;
SELECT MIN(marks) AS Lowest_Marks FROM Exam_Marks;
```





Observation

Execute the SQL statements in MySQL 8.0 Command Line Client. Verify that each statement runs successfully and that the displayed rows match the inserted sample data and query conditions.

Note: Expected aggregate values for the sample data: COUNT=5, SUM=390, AVG=78, MAX=90, MIN=65.

Result

Thus, Experiment 8 (BETWEEN, IN and Aggregate Functions) was performed using MySQL 8.0.